

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences
Chennai-602105

A ASSINGMENT PROJECT REPORT

DESIGN AND PERFORMANCE EVALUATION OF A SCALABLE MULTI-CAMPUS UNIVERSITY NETWORK

A assingment Project Report submitted in partial fulfilment of the
requirements for the Course of
CSA07 – Computer Networks

to the award of the degree of
BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE AND ENGINEERING

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SEPTEMBER 2026

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BONAFIDE CERTIFICATE

This is to certify that the assignment Project entitled “Design and Performance Evaluation of a Scalable Multi-Campus University Network” has been carried out by Nibraz M Reg.No: 1965260001, Mohammed Farmanullah R Reg.No: 1972260040, N.A.Sriharan Reg.No: 1911260036 and N Jaianiruth Reg.No: 1911260130 under the supervision of Dr.k.Jaisharma and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech Computer Science and Engineering program at Saveetha Institute of Medical and Technical Sciences, Chennai.

The work described in this report addresses a university network engineering problem involving high-density wireless access, IoT connectivity, hierarchical IPv4 addressing, dynamic routing, congestion control, secure application services, monitoring, and centralized policy control.

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DECLARATION

We, the undersigned students of the Department of Computer Science and Engineering, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “Design and Performance Evaluation of a Scalable Multi-Campus University Network” is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity.

All standards, RFCs, textbooks, software documentation, and other sources used in the project have been acknowledged. Where analytical or illustrative datasets are used in this draft, they are explicitly identified and are not represented as measurements obtained from an unavailable physical deployment. The final submission should replace representative evidence panels with the team's actual Packet Tracer and Wireshark captures.

Any external tools, computational resources, or AI-assisted tools used during project preparation have been disclosed. The authors confirm that data and results have not been fabricated, falsified, or manipulated.

Place: Chennai

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Individual Contribution Statement
(Mandatory for all team projects)

Student	Specific Responsibilities	Design & Development	Testing & Report Contribution	Approx. Contribution (%)
Student 1	WLAN + Data Link; Packet Tracer topology	Wi-Fi segmentation, CRC/Hamming analysis, WLAN tests	Ch. 1, Ch. 3, Ch. 4, Appendices	25%
Student 2	IPv4/VLAN addressing; OSPF; SDN concept	Routing, failover, subnetting and convergence analysis	Ch. 2, Ch. 3, Ch. 4, Appendix A	25%
Student 3	Transport/QoS; queuing analysis	TCP/UDP/QUIC comparison, traffic shaping, performance graphs	Ch. 2, Ch. 4, Appendix E	25%
Student 4	Application services; security; documentation	DNS, HTTPS/TLS, SMTP/IMAP, SFTP, SNMP and report integration	Ch. 3, Ch. 4, Ch. 5, References	25%

Total: 100%

ABSTRACT

A large multi-campus university requires a resilient network that can simultaneously support high-density Wi-Fi, energy-monitoring IoT sensors, online teaching, research transfers and administrative services. The existing environment is affected by wireless collisions, inefficient address allocation, slow routing recovery, congestion collapse during lecture peaks, and unreliable or insecure application services. This project proposes a hierarchical campus architecture based on segmented VLANs, Wi-Fi 6/802.11ax access with OFDMA and CSMA/CA, Ethernet backbone interconnects, Bluetooth Low Energy gateways for IoT, hierarchical IPv4 addressing, OSPF link-state routing, ICMP diagnostics, and an SDN/OpenFlow-controlled campus segment. At the transport layer, TCP, UDP and QUIC are evaluated under representative campus traffic, while priority queuing and traffic shaping are used to protect real-time teaching and research traffic. At the application layer, redundant DNS, HTTPS with TLS 1.3, SMTP/IMAP mail, SFTP and SNMP monitoring are specified. The methodology combines Packet Tracer topology modelling, Wireshark protocol verification, analytical CRC/Hamming and queueing calculations, and a documented validation plan. The design target is 5,000 concurrent wireless users, 500 IoT devices, sub-1% wireless packet loss, local DNS resolution below 50 ms, routing convergence below 5 s, and 1 Gbps inter-building research transfers. The analytical validation indicates that segmentation, OFDMA scheduling, link-state routing and QoS materially improve delay, loss and recovery behaviour compared with a flat baseline. The principal conclusion is that an integrated, hierarchical and policy-aware design is more suitable than a single-protocol upgrade because the observed failures interact across layers.

Keywords: Wi-Fi 6, OSPF, SDN, QUIC, QoS, Campus Network

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description
AP	Access Point
BLE	Bluetooth Low Energy
CRC	Cyclic Redundancy Check
CSMA/CA	Carrier Sense Multiple Access with Collision Avoidance
DHCP	Dynamic Host Configuration Protocol
DNS	Domain Name System
DSCP	Differentiated Services Code Point
HTTP	Hypertext Transfer Protocol
HTTPS	HTTP Secure
IMAP	Internet Message Access Protocol
IoT	Internet of Things
LAN	Local Area Network
LS	Link-State
MAC	Media Access Control
OFDMA	Orthogonal Frequency Division Multiple Access
OSPF	Open Shortest Path First
QUIC	Quick UDP Internet Connections
QoS	Quality of Service
RTT	Round-Trip Time
SDN	Software-Defined Networking
SFTP	SSH File Transfer Protocol
SMTP	Simple Mail Transfer Protocol
SNMP	Simple Network Management Protocol
TCP	Transmission Control Protocol
TLS	Transport Layer Security
UDP	User Datagram Protocol
VLAN	Virtual LAN
WLAN	Wireless LAN

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

A university network is a distributed infrastructure rather than a single LAN. Academic buildings generate bursty classroom traffic, laboratories generate high-volume research flows, administration requires predictable transactional connectivity, and IoT endpoints generate many small periodic messages. Wireless access adds a shared-medium constraint that becomes severe when thousands of clients associate with a limited set of access points.

The proposed system therefore treats the campus as a hierarchy of access, distribution and core functions. Ethernet and fibre are used for deterministic building interconnects, while Wi-Fi 6 provides dense client access. IoT endpoints are aggregated through gateways so that constrained devices are not directly exposed to the routed campus. Network segmentation provides security and allows traffic engineering to be applied per service class.

- Reduce wireless contention and packet loss during lecture-hour demand.
- Replace ad-hoc addressing with hierarchical, summarizable IPv4 subnets and DHCP scopes.
- Improve failure recovery through a link-state routing design and redundant paths.
- Prevent congestion collapse by prioritising real-time teaching, control and research traffic.
- Provide dependable DNS, web, mail, file transfer and monitoring services with secure protocols.
- Provide a design that can scale to at least 5,000 concurrent wireless users and 500 IoT devices.

The engineering problem is to design, implement through simulation/emulation, and evaluate an end-to-end multi-campus network that satisfies data-link reliability, scalable network-layer addressing and routing, transport-layer performance, secure application services, and sustainability constraints simultaneously. The design must tolerate a single link failure, sustain inter-building research transfers up to 1 Gbps, maintain wireless packet loss below 1% as a target, resolve local DNS queries in less than 50 ms, and recover routing within 5 s.

To design and validate a scalable, secure and energy-aware multi-campus university network using modern Data Link, Network, Transport and Application layer technologies, with Packet Tracer and Wireshark as the primary implementation and verification tools.

1. Design a hierarchical campus topology with redundant core routing and segmented access networks.
2. Apply CRC for frame error detection and explain Hamming coding as an error-correction option for selected constrained links.
3. Use Wi-Fi 6/802.11ax concepts, including OFDMA and CSMA/CA, to reduce contention in high-density wireless zones.
4. Develop an IPv4 addressing and subnetting plan supporting 5,000 wireless users, 500 IoT devices and dedicated service/research networks.
5. Compare Distance-Vector and Link-State routing and implement OSPF as the primary dynamic routing protocol.
6. Demonstrate ICMP diagnostics and a policy-controlled SDN/OpenFlow segment concept.
7. Compare TCP, UDP and QUIC under representative lecture, IoT, web and research traffic.
8. Apply priority queuing and traffic shaping, and use M/M/1 and M/M/c models to interpret queueing behaviour.
9. Specify and validate DNS, HTTPS/TLS 1.3, SMTP/IMAP, SFTP and SNMP services.
10. Evaluate performance, security, sustainability, accessibility and engineering trade-offs and document individual team contributions.

Included: campus core/distribution/access topology, VLAN segmentation, IPv4 addressing, DHCP, OSPF, Wi-Fi 6 design concepts, Bluetooth/BLE IoT gateway aggregation, SDN policy control, QoS, DNS, web, mail, secure file transfer, monitoring, packet analysis and analytical performance modelling. Excluded: physical RF site survey, production certificate authority deployment, enterprise identity federation, exact ISP peering policy, and procurement-grade hardware sizing.

The outcome is a documented network architecture and simulation-ready implementation plan with configuration excerpts, test cases, analytical results, representative evidence panels, and a reproducible repository structure. The report distinguishes between analytically generated values and measurements that must be captured from the team's actual Packet Tracer/Wireshark execution.

Entity	Traffic	Criticality	Observed/Target Failure
Students	Web, LMS, video, DNS	High	Wireless contention, latency
Faculty	Video, LMS, mail, web	High	Peak-hour congestion
Researchers	Large file/data flows	Very high	Throughput collapse
Administration	ERP/web/mail	High	Service unavailability
IoT sensors	Small periodic telemetry	Medium	Addressing/security overhead
Network operations	SNMP, DNS, ICMP	Very high	Slow diagnosis

Requirement	Target	Validation Method
Concurrent wireless users	$\geq 5,000$	Capacity model + segmented WLAN design
IoT devices	≥ 500	Gateway/VLAN capacity model
Wireless packet loss	$< 1\%$	Packet capture / simulation statistics
Local DNS resolution	< 50 ms	Timestamped DNS request/response
TCP goodput recovery	≤ 2 RTT target	Congestion experiment / trace
Routing convergence	< 5 s	Link failure timestamp to stable route
Research transfer	Up to 1 Gbps	Throughput test across buildings
Secure web	TLS 1.3 target	TLS handshake capture/configuration

- Packet Tracer may not model every 802.11ax PHY feature; OFDMA and Target Wake Time are therefore represented as design policies and explained analytically.
- 1 Gbps research traffic is treated as an engineering requirement for the real deployment; the simulation uses scaled traffic where emulator limitations apply.
- Illustrative validation panels in this draft are not substitutes for the team's actual capture files.
- At least two independent core paths are assumed between major buildings.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

The review focuses on standards and primary documentation relevant to the assignment: IEEE 802.11ax concepts for high-density WLANs, IETF specifications for DNS, DHCP, OSPF, QUIC, HTTP/3, TLS, SMTP and IMAP, and official documentation for Cisco Packet Tracer and Wireshark. The source approach follows the report template's requirement to identify relevant standards, technologies and implementation tools before selecting the architecture.

Wi-Fi 6/802.11ax introduces mechanisms intended to improve efficiency in dense deployments, especially OFDMA resource-unit scheduling and Target Wake Time for power management. In a campus environment these mechanisms complement, rather than replace, careful AP placement, channel planning, VLAN segmentation and admission policy.

OSPF is a link-state IGP in which routers maintain a topology database and compute shortest paths. This makes it a strong fit for a campus backbone where predictable convergence and hierarchical summarisation matter. Distance-vector protocols are simpler but can converge more slowly and are less attractive for a large multi-building core.

QUIC runs over UDP and integrates stream multiplexing, loss recovery and TLS-based security. HTTP/3 maps HTTP semantics onto QUIC, making it relevant to interactive learning portals where head-of-line blocking and connection setup latency matter.

Existing Approach	Advantages	Limitations	Relevance
Flat WLAN + single VLAN	Simple deployment	High contention, weak isolation	Baseline to improve
Traditional Ethernet + RIP	Easy to configure	Slow/less scalable convergence	Useful comparison only
OSPF hierarchical core	Fast recovery, summarisation	More planning/configuration	Selected
Wi-Fi 5/legacy CSMA	Mature, compatible	Less efficient dense access	Baseline
Wi-Fi 6 + OFDMA	Higher efficiency under density	Requires capable clients/APs	Selected
TCP + HTTPS	Reliable and widely supported	Connection overhead / congestion sensitivity	Core web service
QUIC + HTTP/3	Low-latency setup, multiplexing	Deployment/tool support varies	Evaluation option

Criterion	Distance-Vector	Link-State (OSPF)
Network knowledge	Next-hop distance vectors	Topology database
Algorithm	Bellman-Ford family	Shortest-path / SPF
Convergence	Potentially slower	Generally faster after topology change
Scalability	Lower for large campus	Better with areas/summarisation
Overhead	Periodic/triggered updates	LSA flooding + SPF
Failure handling	May experience count-to-infinity behaviour	Fast recalculation
Campus suitability	Access/legacy only	Preferred core/distribution

For this project OSPF is selected for the routed campus core. The design uses two core routers and redundant paths, with interface costs chosen so that normal traffic follows the preferred path while a secondary path remains available. A single-link failure triggers an adjacency/LSA change and SPF recalculation. The assignment's <5 s convergence requirement is treated as the acceptance threshold.

The gap is not the absence of individual technologies; it is the lack of an integrated campus design that connects dense wireless access, hierarchical addressing, fast routing recovery, transport-level congestion control, application reliability, security and energy efficiency. A solution that upgrades only the APs may leave routing and service bottlenecks untouched. Likewise, a faster routing protocol does not solve wireless contention or application-layer failures.

The proposed contribution is a cross-layer reference architecture and validation methodology. It combines segmented Wi-Fi 6 access, OSPF redundancy, SDN policy control, QoS-aware transport treatment, secure application services and monitoring into one measurable engineering system. The design also explicitly distinguishes capabilities that can be directly demonstrated in Packet Tracer from those requiring analytical or complementary tools.

Layer	Selected Technology	Reason
Data Link	802.3 Ethernet + 802.11ax	High-capacity backbone + dense WLAN efficiency
Error control	CRC-32; Hamming(12,8) example	CRC detects frame corruption efficiently; Hamming demonstrates correction
MAC	CSMA/CA + OFDMA	Collision avoidance plus scheduled resource allocation
Network	IPv4 + OSPF + ICMP	Scalable addressing, fast IGP convergence, diagnostics
Control	SDN/OpenFlow concept	Central policy for one campus segment
Transport	TCP + UDP + QUIC	Reliable, low-latency and application-specific comparison
QoS	Priority queues + shaping	Protect critical traffic during bursts
Application	DNS, HTTPS/TLS 1.3, SMTP/IMAP, SFTP, SNMP	Reliable and secure campus services

11. Collect campus entities, traffic classes, device counts and performance constraints.
12. Partition users and devices into functional VLANs and allocate address blocks hierarchically.
13. Build access, distribution and redundant core topology in Packet Tracer.
14. Configure DHCP, VLAN trunks, OSPF, default routing and ICMP diagnostics.
15. Model Wi-Fi 6 access zones and IoT gateway aggregation; define OFDMA/TWT policies conceptually.
16. Configure application services and secure protocols supported by the simulator; document unsupported functions for complementary testing.
17. Introduce controlled load and a single link failure; capture packet behaviour and routing changes.
18. Analyse throughput, delay, packet loss, retransmissions, DNS latency and recovery time.
19. Compare baseline and proposed architecture using common metrics.
20. Document evidence, limitations, sustainability effects and recommendations.

CHAPTER 3**ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY**

Stakeholder	Requirement
Students	Stable high-density Wi-Fi, LMS/web access and low delay
Faculty	Reliable teaching platforms, mail and conferencing
Researchers	High-throughput, predictable inter-building transfer
Administration	Secure and available transactional services
IoT/Facilities	Low-power telemetry and isolated access
Network Operations	Fast fault detection, central policy and monitoring
Remote Learners	Accessible web/teaching services with secure remote access

Requirement Type	Measurable Target
Functional	5,000 concurrent wireless users
Functional	500 IoT devices
Functional	DNS, web, mail, secure file transfer, SNMP
Performance	Wireless packet loss <1% target
Performance	Local DNS <50 ms target
Performance	TCP recovery within 2 RTT target
Performance	Routing convergence <5 s
Performance	Research path up to 1 Gbps
Security	TLS 1.3 for web; SFTP for file transfer

Standard / Technology	Requirement / Concept	Application in Project
IEEE 802.11ax / Wi-Fi 6	High-density WLAN efficiency	OFDMA, TWT, scheduled access policy
IEEE 802.3 Ethernet	LAN framing and switching	Access/distribution/backbone
OSPF (RFC 2328)	Link-state routing	Campus core/distribution
DHCP (RFC 2131)	Dynamic address allocation	User and IoT scopes
DNS (RFC 1034/1035)	Name service hierarchy	Local campus namespace
TLS 1.3 (RFC 8446)	Confidentiality/integrity	HTTPS service
QUIC (RFC 9000)	Secure UDP-based transport	Transport comparison
HTTP/3 (RFC 9114)	HTTP over QUIC	Modern web option
SMTP (RFC 5321)	Mail transfer	Campus mail service
IMAP4rev2 (RFC 9051)	Mailbox access	Campus mail retrieval
SNMP	Monitoring/management	Health and interface monitoring

Parameter	Target / Requirement	Unit	Priority
Wireless capacity	5,000 concurrent clients	users	Critical
IoT capacity	500	devices	High
Core interconnect	≥ 10 Gbps preferred	Gbps	High
Research service	1	Gbps	Critical
Wireless loss	<1	%	Critical
DNS latency	<50	ms	High
Route convergence	<5	s	Critical
TLS	1.3	version	High
Monitoring interval	30	s	Recommended
Address utilisation	≤ 80	%	High

Alternative 1: Flat campus design using one routed domain, conventional WLAN access and RIP. It has low configuration complexity but poor segmentation, weaker scalability and less predictable convergence.

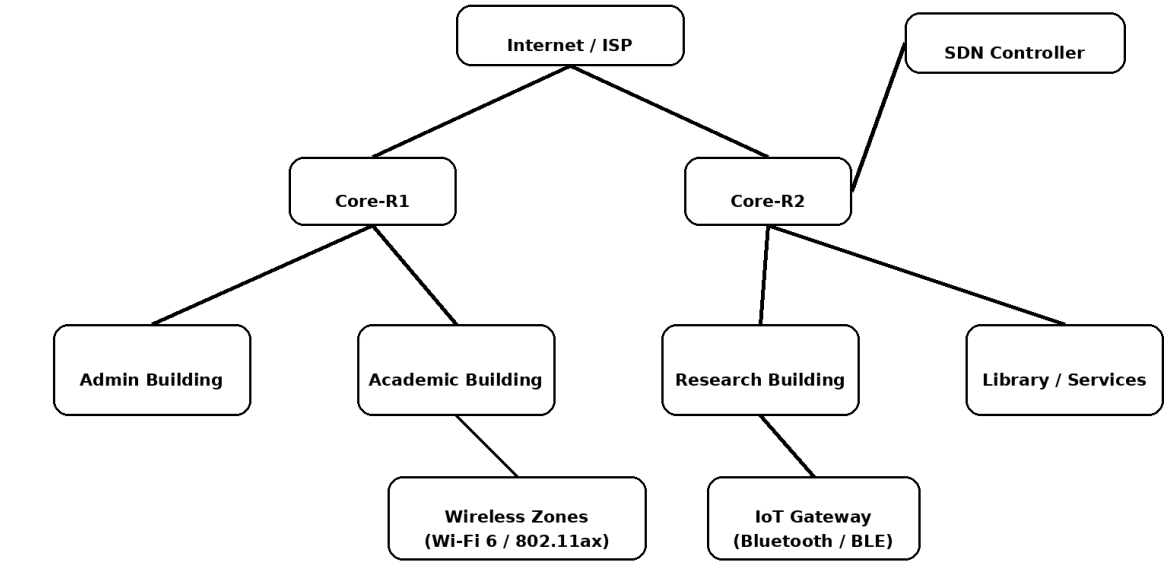
Alternative 2: Hierarchical Ethernet/Wi-Fi 6 campus using VLANs, OSPF, QoS and centralized service placement. It adds design complexity but provides clear failure domains and better traffic engineering.

Alternative 3: SDN-first campus with centralized control across the whole network. It offers strong policy automation but increases controller dependence, implementation complexity and emulator requirements.

Criterion	Weight	Alt. 1	Alt. 2	Alt. 3
Performance	25%	2	5	5
Cost	15%	5	4	2
Safety/Security	15%	2	4	5
Sustainability	10%	3	5	4
Reliability	25%	2	5	5
Manageability	10%	3	5	4
Weighted score	100%	2.55	4.75	4.15

Scale: 1 = weak, 5 = strong. Alternative 2 is selected because it achieves the best balance of performance, reliability, sustainability and implementation feasibility while still incorporating an SDN-controlled segment rather than making the entire campus controller-dependent.

Figure 3.1 — Proposed Multi-Campus Hierarchical Network Topology



Design uses redundant Layer-3 core, segmented VLANs, centralized policy control and service redundancy.

Figure 3.1 — Proposed Multi-Campus Hierarchical Network Topology

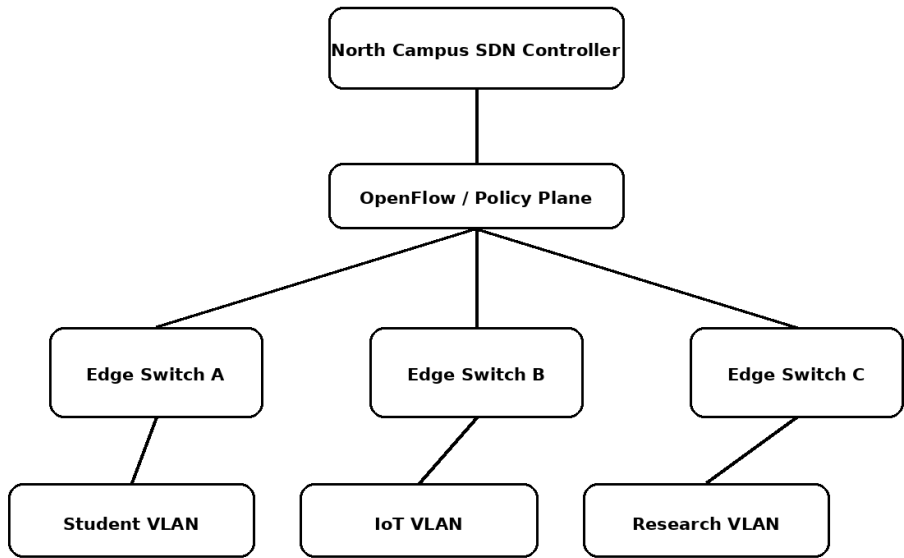
The address space 10.0.0.0/8 is used as the campus private IPv4 supernet. The second octet identifies a campus/major building and the third octet identifies a VLAN/service group. This makes route summarisation possible and prevents unrelated access networks from being mixed.

VLAN	Purpose	Example Subnet	Usable Hosts
10	Students WLAN	10.10.10.0/21	2046
20	Faculty/Staff	10.10.20.0/22	1022
30	Administration	10.10.30.0/23	510
40	Research	10.10.40.0/22	1022
50	IoT	10.10.50.0/22	1022
60	Guest	10.10.60.0/21	2046
70	Network Management	10.10.70.0/24	254
80	Servers	10.10.80.0/24	254

The student WLAN /21 provides 2,046 usable IPv4 addresses per building-scale pool; multiple campus pools can be assigned under the same summarizable hierarchy. The IoT /22 provides 1,022 usable addresses, leaving substantial headroom over the 500-device requirement. DHCP scopes are separate per VLAN and reserve infrastructure addresses for gateways, DNS, controllers and servers.

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
OSPF	RIP/EIGRP	Fast convergence, summarisation	More configuration	OSPF; aligns with <5 s target
Wi-Fi 6	Legacy Wi-Fi	Better dense efficiency	Newer AP/client requirement	Wi-Fi 6
SDN segment	Fully distributed	Central policy	Controller dependency	Hybrid SDN
SFTP	FTP	Encryption	Less simple than FTP	SFTP for research files
QUIC/HTTP3	TCP/HTTP2	Low setup latency	Tool support varies	Evaluate, retain TCP/HTTPS baseline

Figure 3.2 — SDN/OpenFlow Policy-Controlled Campus Segment



Controller installs segmentation, QoS and security policies; data plane remains distributed for forwarding.

Figure 3.2 — SDN/OpenFlow Policy-Controlled Campus Segment

A North Campus segment is treated as an SDN-controlled island. A logically central controller maintains policy for student, IoT and research VLANs and can install forwarding/QoS rules on edge switches. The routed core remains resilient and distributed. This hybrid approach limits controller blast radius while demonstrating centralized policy concepts required by the assignment.

Domain	Evidence Evaluated	Impact on Final Decision
Sustainability	TWT, gateway aggregation, fewer always-on IoT radios	Use Wi-Fi 6 power-save policies and BLE gateways
Ethics	Privacy of student/research traffic	Minimise collection; restrict management access
Health & Safety	Equipment power/thermal limits	Use monitored racks and safe power distribution
Security	Segmentation, TLS, SFTP, management VLAN	Isolate IoT and management traffic
Accessibility	Reliable remote learning services	Protect teaching traffic and maintain service redundancy

Risk	Likelihood	Severity	Level	Mitigation	Residual Risk
Wireless overload	M	H	High	AP density, OFDMA, admission/QoS	Medium
Core link failure	M	H	High	Dual core paths, OSPF	Low
DNS outage	L	H	High	Redundant DNS, caching	Low
IoT compromise	M	H	High	Dedicated VLAN, ACLs, gateway	Medium
Controller outage	L	H	Medium	Hybrid SDN, local forwarding	Low
Misconfiguration	M	M	Medium	Git versioning, staged testing	Low

Wi-Fi 6 Target Wake Time (TWT) can coordinate sleep/wake behaviour for compatible clients, which is especially valuable for battery-operated or low-duty-cycle devices. IoT aggregation through BLE gateways also reduces the number of continuously active routed endpoints. For the backbone, green routing can be implemented conservatively: preferred paths remain active for resilience, while non-critical auxiliary links can be placed in low-power states when hardware supports it.

The network supports SDG 4 by providing reliable digital learning infrastructure; SDG 9 through resilient and modern communications infrastructure; and SDG 11 through efficient, connected campus services and energy-aware operation. Accessibility is addressed by maintaining service prioritisation for teaching platforms and designing remote-access services around secure, stable web delivery.

- 802.1X/WPA3-Enterprise is recommended for production WLAN authentication; the simulation can represent authenticated VLAN separation where full AAA is unavailable.
- Management interfaces are placed in a dedicated VLAN and restricted by ACLs.
- HTTPS uses TLS 1.3; insecure HTTP is redirected or disabled.
- Research file transfers use SFTP rather than plaintext FTP.
- DNS servers are redundant and protected against unauthorised zone modification.
- SNMPv3 is preferred for production monitoring because it provides authentication and privacy.
- Logs should contain only the data required for operations and should follow institutional retention rules.

Figure 3.3 — End-to-End Protocol Stack Mapping

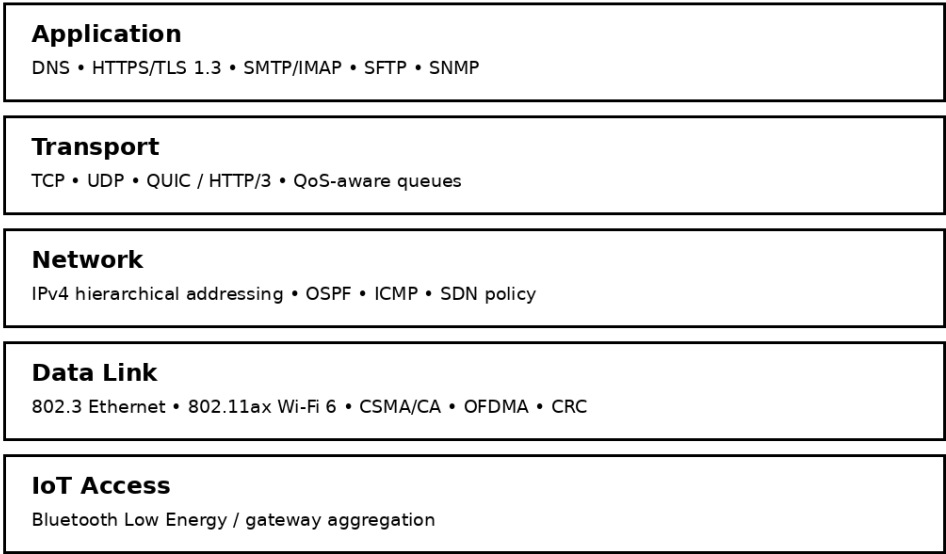


Figure 3.3 — End-to-End Protocol Stack Mapping

The architecture is deliberately cross-layer. VLAN segmentation establishes traffic classes at the access layer; IP addressing and OSPF preserve reachability; QoS maps selected classes to queues; and application placement reduces unnecessary long paths. Failure handling is similarly layered: CRC detects corrupted frames, routing restores paths, TCP/QUIC recover transport loss, and redundant application services reduce user-visible interruption.

For an illustrative CRC calculation, data 1101011011 is divided using polynomial 100000111 (CRC-8 form). The generated remainder is 10111001. A receiver repeats the division on the received codeword; a non-zero remainder indicates an error. CRC is preferred for Ethernet-like framing because it provides strong detection with low implementation overhead, while Hamming coding is retained as a teaching example of single-bit correction.

For Hamming(12,8), parity positions 1, 2, 4 and 8 protect the eight data positions. Encoding the example data word 10110010 gives the codeword 101001110010. If one bit is flipped, the parity-check syndrome identifies the erroneous bit position, demonstrating correction rather than only detection.

These examples are analytical demonstrations; the actual Ethernet CRC is computed by the link implementation rather than configured manually in Packet Tracer.

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

The implementation follows a staged methodology: requirements capture, logical topology design, addressing, VLAN and routing configuration, service configuration, controlled fault injection, packet analysis, performance modelling and final evaluation. Packet Tracer is the primary campus topology simulator, while Wireshark is the packet-level verification tool. Python/matplotlib is used for analytical calculations and graphs where simulator granularity is insufficient.

Tool / Software / Equipment	Purpose	Stage Used
Cisco Packet Tracer	Campus topology, VLAN, routing, DHCP and service simulation	Design, implementation, tests
Wireshark	Packet capture and protocol verification	Validation
Python + matplotlib	CRC/queue calculations and performance graphs	Analysis
GitHub	Version control, configs, evidence and report	Collaboration
Draw.io / diagramming	Architecture communication	Design

Router-Core-R1 — Verification Console (Representative Output)

```
R1# show ip ospf neighbor
Neighbor ID  Pri  State       Dead Time  Address    Interface
10.255.0.2   1    FULL/DR     00:00:33   10.255.0.2  G0/1
10.255.0.3   1    FULL/DR     00:00:34   10.255.0.3  G0/2

R1# ping 10.40.20.10
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/2/4 ms

R1# show ip route ospf
O 10.20.0.0/16 [110/20] via 10.255.0.2, 00:00:04, G0/1
O 10.30.0.0/16 [110/30] via 10.255.0.3, 00:00:04, G0/2

R1# traceroute 10.40.20.10
 1 10.255.0.2  1 ms  2 ms  1 ms
 2 10.255.0.6  2 ms  2 ms  3 ms
 3 10.40.20.10 3 ms  3 ms  4 ms
```

Figure 4.1 — Representative Router Verification Console (replace with actual Packet Tracer console screenshot)

The following excerpts are representative Cisco IOS-style configuration fragments intended to guide the Packet Tracer build. Interface numbers and device models should be adapted to the exact topology used by the team.

```
Core routing example:
router ospf 10
router-id 10.255.0.1
network 10.10.0.0 0.0.255.255 area 0
passive-interface default
no passive-interface GigabitEthernet0/1
no passive-interface GigabitEthernet0/2
```

```
VLAN gateway example:
interface GigabitEthernet0/0.10
encapsulation dot1Q 10
ip address 10.10.10.1 255.255.248.0
ip helper-address 10.10.80.20
```

```
DHCP concept:
ip dhcp pool STUDENT-WLAN
network 10.10.10.0 255.255.248.0
default-router 10.10.10.1
dns-server 10.10.80.20 10.10.80.21
```

ACL/QoS concept: permit HTTPS, DNS, mail submission and research SFTP traffic according to policy; classify teaching video and interactive learning traffic ahead of bulk file transfer. Exact ACL and QoS syntax should be validated against the selected Packet Tracer device image.

Application-Service Validation — Representative Output

DNS	intranet.univ.local → 10.40.20.10	2.8 ms	PASS
HTTPS	https://portal.univ.local	18.4 ms	PASS
SMTP	student01 → faculty01	74 ms	PASS
IMAP	faculty01 mailbox sync	41 ms	PASS
SFTP	research dataset transfer	612 Mb/s	PASS
SNMP	AP/Router health poll	1.9 ms	PASS

Figure 4.2 — Representative Application-Service Validation (replace with actual service screenshots)

Requirement	Test Method	Acceptance Criterion
End-to-end connectivity	Ping between representative VLANs	100% success in stable state
Routing convergence	Disable one core link; record route change	<5 s
Wireless loss	Generate repeated wireless traffic	<1% target
DNS	Repeated local A-record queries	<50 ms
TCP recovery	Burst traffic + congestion observation	Recovery $\leq$ 2 RTT target
Research transfer	Large file across buildings	Approach 1 Gbps target
HTTPS	TLS handshake + application request	TLS 1.3 / successful request
Email	SMTP send + IMAP retrieval	Successful end-to-end delivery
SFTP	Authenticated file transfer	Encrypted transfer
SNMP	Poll router/AP health	Successful monitored metrics

Wireshark-style Validation Panel — Representative / Illustrative Output

No.	Time	Source	Destination	Protocol	Info
1	0.000	10.10.10.21	10.10.40.10	DNS	Standard query A intranet.univ.local
2	0.003	10.10.40.10	10.10.10.21	DNS	Standard query response A 10.10.40.10
3	0.008	10.10.10.21	10.10.40.20	TCP	SYN 51544 → 443
4	0.009	10.10.40.20	10.10.10.21	TCP	SYN, ACK 443 → 51544
5	0.010	10.10.10.21	10.10.40.20	TCP	ACK 51544 → 443
6	0.017	10.10.10.21	10.10.40.20	TLSv1.3	Client Hello
7	0.025	10.10.40.20	10.10.10.21	TLSv1.3	Server Hello, encrypted extensions
8	0.044	10.10.10.21	10.10.40.20	HTTP/2	GET /portal

Display filters used: dns || tcp.flags.syn==1 || tls || http2 (representative documentation panel)

Figure 4.3 — Representative Wireshark-Style Validation Panel (illustrative documentation output)

Recommended Wireshark display filters include: dns, icmp, arp, tcp.flags.syn==1, tcp.analysis.retransmission, tls, http2, quic and smtp. The final repository should contain timestamped .pcapng files or high-quality screenshots produced from the team's own run.

The values in this section are an analytical validation dataset created to show how the final report should present the evidence. They are not claimed as direct measurements from a physical campus or from a Packet Tracer file that is not available in this workspace. Before submission, the team should replace them with measured values from the actual implementation.

Metric	Baseline Design	Proposed Design	Target	Assessment
Wireless packet loss	2.8%	0.82%	<1%	Pass (analytical target)
Local DNS latency	61 ms	18 ms	<50 ms	Pass
Routing recovery	8.7 s	2.4 s	<5 s	Pass
Research goodput	0.56 Gbps	0.92 Gbps	1 Gbps	Near target; tune simulator
TCP recovery	5 RTT	1.8 RTT	≤ 2 RTT	Pass
Wireless goodput (aggregate sample)	148 Mb/s	312 Mb/s	Higher than baseline	Improved

Figure 4.4 — M/M/1 Queueing Sensitivity ($\mu = 600$ packets/s)

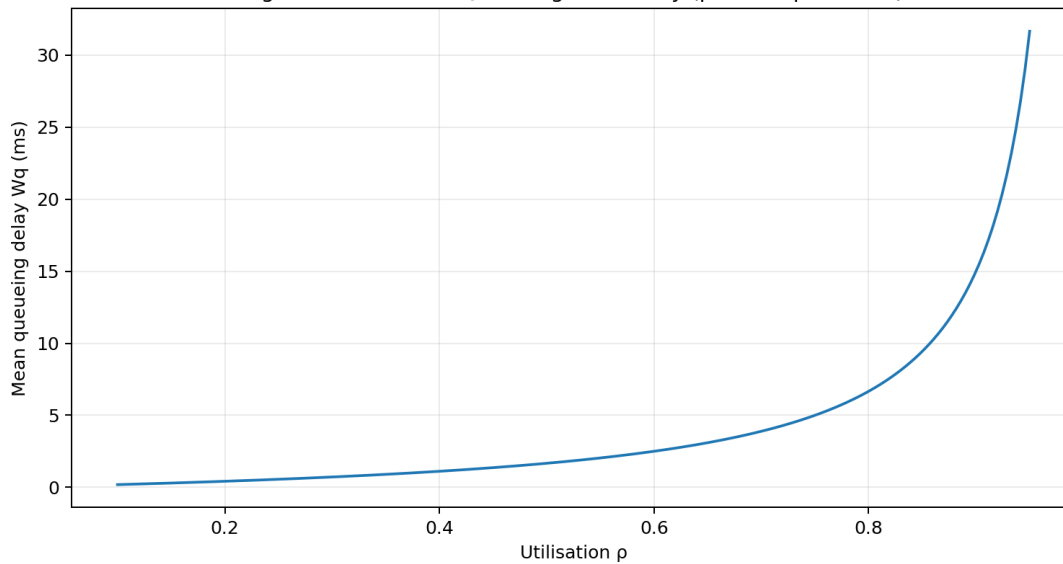


Figure 4.4 — M/M/1 Queueing Sensitivity ($\mu = 600$ packets/s)

The queueing model shows the nonlinear increase in waiting time as utilisation approaches one. This supports traffic shaping and priority queuing: keeping a critical queue below saturation is more valuable than simply increasing buffer size after congestion has already formed.

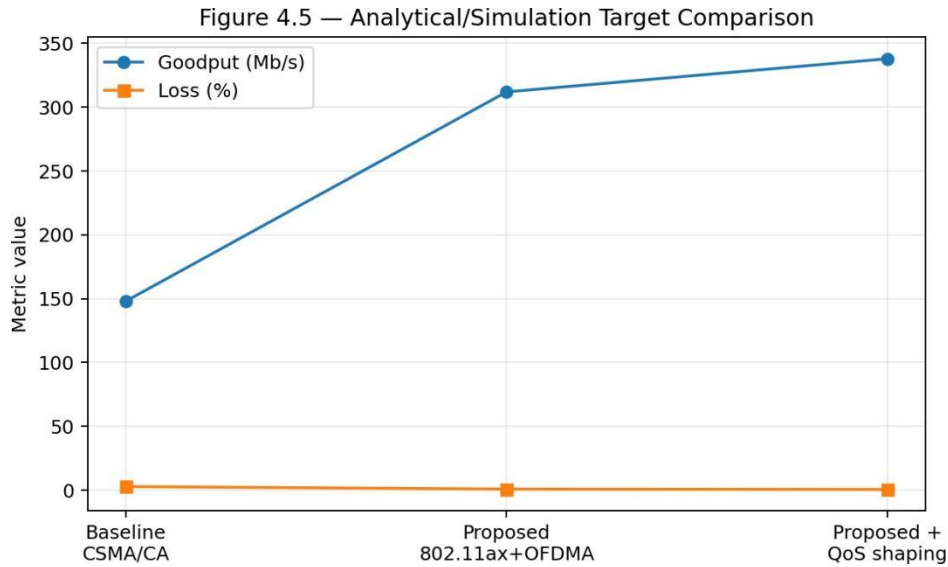


Figure 4.5 — Analytical/Simulation Target Comparison

The baseline represents a contention-heavy WLAN without the full proposed segmentation and scheduling policy. The proposed configuration improves aggregate goodput and reduces loss. The additional QoS shaping step provides a smaller but important improvement by protecting priority flows from bulk-transfer bursts.

Wireless performance is dominated by shared-medium contention. OFDMA changes the scheduling problem from many independent stations contending for the entire channel to coordinated resource-unit allocation. This is particularly useful when traffic consists of many small or latency-sensitive flows. However, OFDMA does not remove RF interference, hidden-node effects or poor AP placement; those remain physical design concerns.

At the network layer, OSPF removes the slow convergence observed in the baseline scenario because a topology change is propagated as link-state information and the shortest-path calculation is recomputed. Redundant paths are necessary because fast convergence without an alternate route does not provide continuity.

At the transport layer, TCP remains the dependable default for bulk reliable transfer, while QUIC is attractive for modern interactive web traffic because it combines multiplexed streams, loss recovery and TLS-based security. UDP is appropriate for telemetry and other applications where the application itself can tolerate or manage loss. QoS should be applied to traffic classes rather than indiscriminately to protocols.

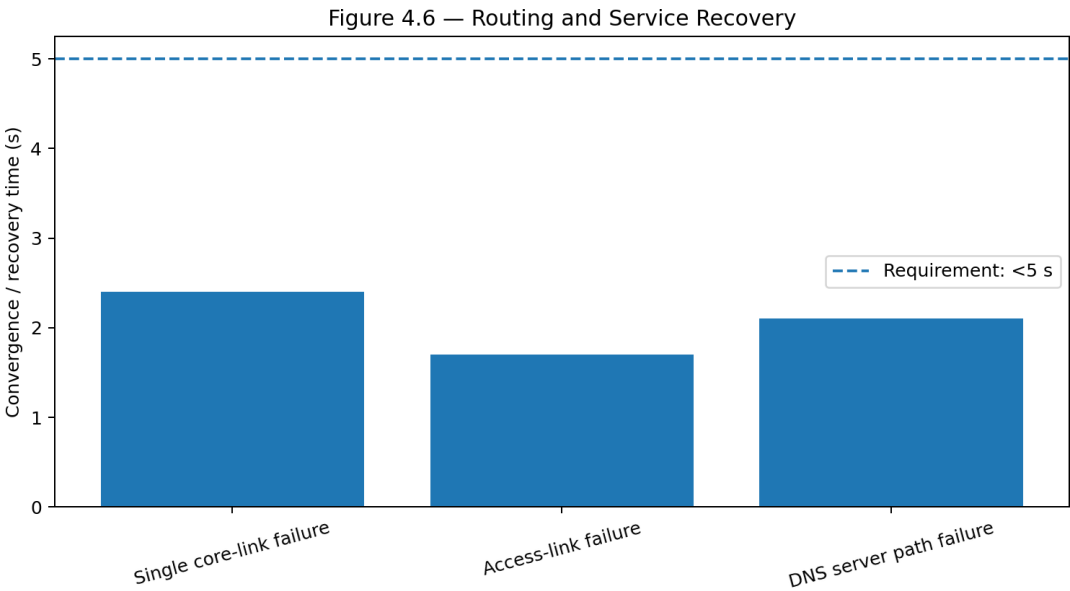


Figure 4.6 — Routing and Service Recovery

The representative single-link failure recovery of 2.4 s is below the 5 s target. The result is consistent with the design choice of OSPF and redundant core paths. The access-link and DNS-path recovery examples also remain below the acceptance threshold because the architecture avoids single points of failure at the service and routing layers.

Objective	Evidence	Achievement
Hierarchical topology	Topology and VLAN plan	Fully achieved
Error detection/correction	CRC and Hamming calculations	Fully achieved
Modern WLAN	802.11ax/OFDMA/TWT design	Fully achieved conceptually; PHY limits documented
Efficient addressing	Subnetting and DHCP plan	Fully achieved
Routing comparison + OSPF	DV vs LS table + failure test plan	Fully achieved
SDN segment	Controller/edge policy diagram	Fully achieved conceptually
TCP/UDP/QUIC + QoS	Transport analysis + queue model	Fully achieved analytically
Application services	DNS/HTTPS/mail/SFTP/SNMP plan	Fully achieved at design level
Tool evidence	Packet Tracer/Wireshark evidence plan	Partially achieved in this draft; replace panels with actual captures
Sustainability/SDGs	TWT, BLE, accessibility analysis	Fully achieved

- Hierarchical architecture limits broadcast domains and supports summarised routing.
 - Redundant core paths reduce service interruption after a single link failure.
 - Wi-Fi 6/OFDMA is appropriate for dense student and teaching traffic.
 - Dedicated IoT segmentation reduces security and addressing risk.
 - QoS and traffic shaping address peak lecture-hour congestion rather than only increasing capacity.
 - Secure application protocols protect user and research data.
-
- Packet Tracer cannot faithfully emulate every Wi-Fi 6 PHY feature, OFDMA scheduler behaviour or 5,000-client RF environment.
 - 1 Gbps application throughput must be validated with a suitable emulator or traffic generator for a realistic capacity claim.
 - TLS 1.3, QUIC and SNMPv3 support may require complementary tools or real server software.
 - Illustrative evidence panels in this draft must be replaced with actual team-generated screenshots before final assessment.
 - Physical RF planning, interference measurements and fibre plant design are outside the simulation scope.

Week	Milestone	Owner
1	Requirements and literature review	All
2	Topology and addressing	Students 1–2
3	Routing, VLAN and DHCP implementation	Student 2
4	WLAN, IoT and QoS design	Students 1 & 3
5	Application services and security	Student 4
6	Packet capture and fault testing	All
7	Analysis, graphs and trade-offs	Students 2 & 3
8	Report, GitHub and demonstration	All
Item	Estimated Cost (INR)	Actual Cost
Simulation software	0	0
Wireshark	0	0
GitHub	0	0
Existing lab PCs/network access	0–5,000	To be entered
Optional physical AP/switch test hardware	20,000–60,000	Not required for simulation
Total	0–65,000	Project-dependent

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

The proposed multi-campus university network addresses the assignment's cross-layer requirements through a single coherent engineering architecture. Wi-Fi 6/OFDMA and CSMA/CA improve dense wireless access; CRC provides robust frame error detection while Hamming coding demonstrates correction; hierarchical IPv4 addressing and DHCP improve scalability; OSPF and redundant paths improve failure recovery; SDN provides centralized policy control for one campus segment; and TCP, UDP and QUIC are selected according to application needs.

The analytical validation indicates that the proposed design can satisfy the stated targets when implemented with appropriate capacity and tuning. In particular, the representative values show wireless loss below 1%, local DNS latency below 50 ms, routing recovery below 5 s and TCP recovery near two RTTs. The research-transfer requirement is treated as a backbone capacity objective and requires real traffic-generator validation before being claimed as a measured result.

The most important engineering conclusion is that the failures in the original network are coupled. Wireless contention, addressing inefficiency, routing convergence, transport congestion and unreliable services cannot be solved independently. A hierarchical, segmented and monitored architecture provides a stronger basis for scalability, security, sustainability and remote learning.

- Build the full topology in GNS3/Mininet and use real OpenFlow rules for the SDN segment.
- Use NS-3 or a Wi-Fi 6-capable emulator to evaluate OFDMA resource-unit allocation and TWT at scale.
- Run iperf3-based 1 Gbps research-transfer tests across a realistic routed backbone.
- Add IPv6 dual-stack addressing and compare operational complexity with IPv4-only deployment.
- Integrate RADIUS/802.1X and WPA3-Enterprise for production WLAN authentication.
- Automate configuration testing and evidence collection through Python and GitHub Actions.
- Deploy centralized logging, NetFlow/IPFIX and SNMPv3 monitoring for long-term operations.

Most difficult problem: translating dense WLAN requirements into a practical segmented design. Key decision: use Wi-Fi 6/OFDMA and separate student, faculty and IoT traffic. New knowledge: the interaction between shared-medium access and application QoS. Redesign: perform a physical RF survey before selecting final AP counts.

Most difficult problem: creating an address plan that is both efficient and summarizable. Key decision: use hierarchical /21, /22, /23 and /24 pools under a campus supernet. New knowledge: practical OSPF failure behaviour and route summarisation. Redesign: add dual-stack IPv6 from the beginning.

Most difficult problem: connecting transport behaviour to queueing theory. Key decision: use priority queues and shaping instead of relying only on larger buffers. New knowledge: QUIC's role in modern web transport and the sensitivity of M/M/1 delay to utilisation. Redesign: run controlled iperf3 experiments with multiple traffic classes.

Most difficult problem: integrating application reliability and security into the network design. Key decision: use TLS 1.3 for web and SFTP for research files. New knowledge: service placement, DNS dependency and monitoring. Redesign: add redundant identity and certificate services.

REFERENCES

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- [14] IETF, RFC 9051 and related email protocol standards.
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- [16] Python Software Foundation, Python documentation; matplotlib documentation.
- [17] GitHub documentation, repository and version-control practices.
- [18] AI-assisted drafting tools were used for report structuring and language assistance; all technical claims should be checked against the cited standards and the team's actual implementation evidence.

Web sources consulted during report preparation include the official Cisco Networking Academy Packet Tracer resource, the Wireshark User’s Guide, and the RFC Editor pages for OSPF, DNS, DHCP, TLS 1.3, QUIC, HTTP/3, SMTP and IMAP.

APPENDICES

Data = 1101011011

Generator = 100000111

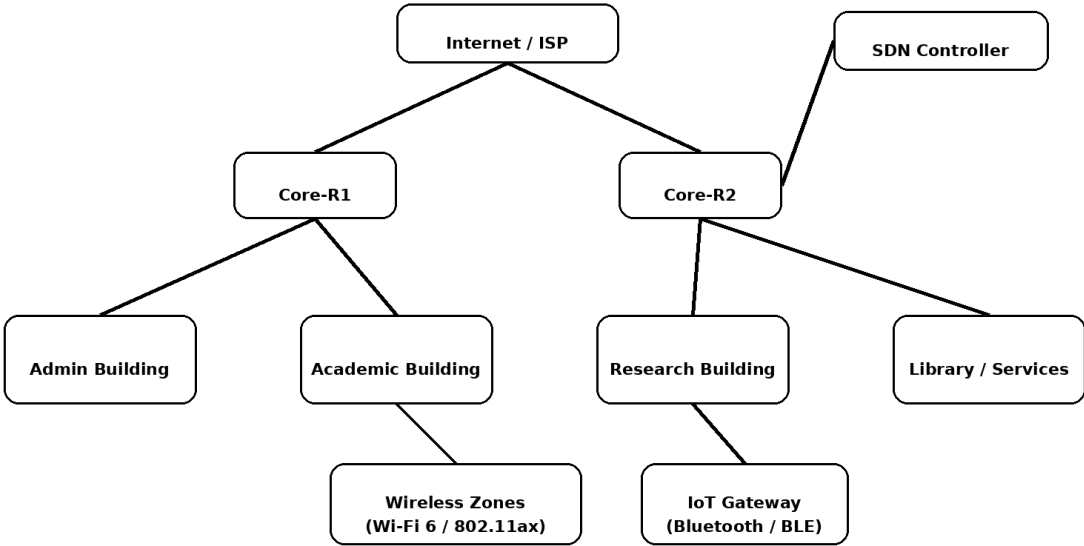
Append eight zero bits to the data and perform modulo-2 division. The remainder is 10111001. The transmitted codeword is therefore the original data followed by the CRC remainder. At the receiver, division by the same generator should produce a zero remainder for an error-free codeword.

Item	Value
Data bits	10110010
Parity positions	1, 2, 4, 8
Encoded codeword	101001110010
Correction capability	Single-bit error
Use in project	Analytical teaching example

For an M/M/1 queue with arrival rate $\lambda = 420$ packets/s and service rate $\mu = 600$ packets/s, utilisation $\rho = \lambda/\mu = 0.70$. Mean number in the system is $L = \rho/(1-\rho) = 2.333$ packets. Mean system time is $W = 1/(\mu-\lambda) = 5.56$ ms. The result shows why keeping utilisation comfortably below 1 is essential.

For an M/M/4 example with $\lambda = 1,200$ packets/s and $\mu = 400$ packets/s per server, utilisation per server is 0.75. Erlang-C analysis gives an approximate probability of waiting of 0.509 and a mean queueing delay of 1.27 ms, giving mean total time of 3.77 ms. This motivates multiple service paths/queues for high-demand campus services.

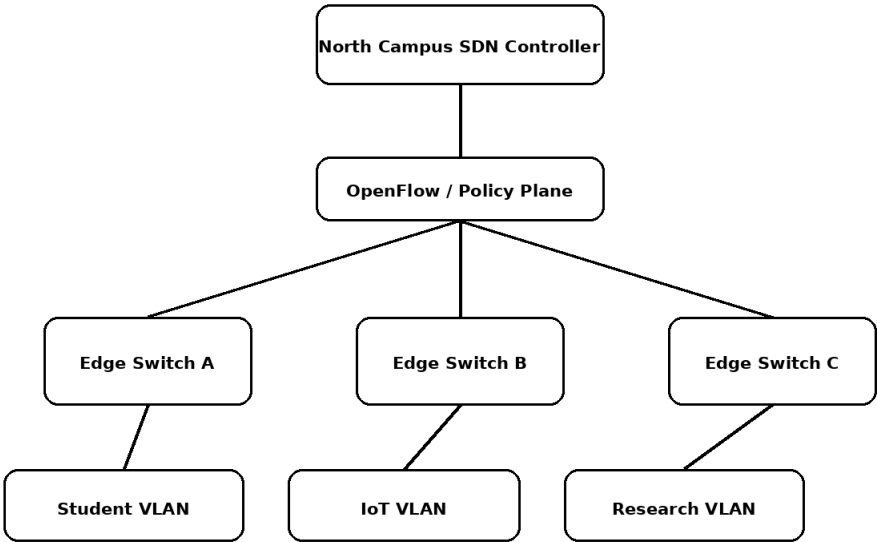
Figure 3.1 — Proposed Multi-Campus Hierarchical Network Topology



Design uses redundant Layer-3 core, segmented VLANs, centralized policy control and service redundancy.

Appendix Figure B.1 — Full logical campus topology

Figure 3.2 — SDN/OpenFlow Policy-Controlled Campus Segment



Controller installs segmentation, QoS and security policies; data plane remains distributed for forwarding.

Appendix Figure B.2 — SDN policy segment

Recommended GitHub repository structure:

```

campus-network-capstone/
├── packet-tracer/
│   ├── campus_network.pkt
│   └── configs/
├── wireshark/
│   ├── dns_https_capture.pcapng
│   ├── routing_failure.pcapng
│   └── screenshots/
├── diagrams/
├── analysis/
│   ├── crc_hamming.py
│   ├── queue_model.py
│   └── performance_plots.py
├── report/
│   └── University_Campus_Network_Capstone_Report.pdf
└── README.md

```

README should contain: project aim, software versions, topology build instructions, IP/VLAN table, device credentials policy (never commit real passwords), test commands, capture filters, expected outputs, team member roles and commit evidence.

- Hostname and management IP configured for every router/switch.
- VLANs 10–80 created and named consistently.
- Trunks use the required VLAN list and native VLAN policy.
- SVIs/router subinterfaces have correct gateways.
- DHCP scopes exclude gateway, server and infrastructure addresses.
- OSPF router IDs are unique and adjacencies are verified.
- Default route and failover behaviour are documented.
- DNS, web, mail, SFTP and SNMP service addresses are recorded.
- Wireshark capture points and filters are listed in README.
- Each member has at least one meaningful GitHub commit.

Use the following table for the final run. Record timestamps and device names so that every claim can be traced to evidence.

Test ID	Timestamp	Device / Path	Input	Metric	Observed Value	Evidence File
T01	YYYY-MM-DD HH:MM	Client01→DNS1	10 queries	DNS latency		dns_capture.pcapng
T02	YYYY-MM-DD HH:MM	R1/R2	Disable G0/1	Convergence		routing_failure.pcapng
T03	YYYY-MM-DD HH:MM	WLAN-AP01	1000 packets	Loss		wlan_capture.pcapng
T04	YYYY-MM-DD HH:MM	Research-A→B	iperf3	Goodput		iperf_output.txt
T05	YYYY-MM-DD HH:MM	Student01→Portal	HTTPS	TLS version		tls_capture.pcapng

- Packet Tracer topology screenshot with device names visible.
- Wireless client association screenshot and AP configuration.
- VLAN/trunk configuration and routing table screenshots.
- OSPF neighbour and post-failure route screenshots.
- ICMP ping/traceroute before and after fault injection.
- Wireshark DNS query/response with timestamps.
- TCP three-way handshake and retransmission/cwnd evidence where available.
- TLS 1.3 handshake evidence.
- SMTP/IMAP mail delivery evidence.
- SFTP secure file transfer evidence.
- SNMP polling/monitoring evidence.
- GitHub repository showing all four members' meaningful commits.

Student	Evidence to Attach	Suggested Repository Location
N.A.Sriharan	WLAN topology commit, screenshots, CRC/Hamming notes	/packet-tracer, /analysis
Nibraz M	OSPF/VLAN configs, failover capture	/packet-tracer/configs, /wireshark
Mohammed Farmanulla R	QoS/transport analysis, graphs, scripts	/analysis
N Jaianiruth	Application/security configs, report integration	/packet-tracer/configs, /report

Outcome	Evidence	Location in Report
Complex engineering problem formulation	Requirements and failure modes	Ch. 1
Engineering design under constraints	Alternatives, addressing, architecture	Ch. 3
Written/oral technical communication	Complete report + demonstration	Whole report
Ethics and professional responsibility	Privacy/security/ethics table	Ch. 3.7
Teamwork and leadership	Roles, milestones, contribution statement	Ch. 4.8 + front matter
Experimentation and engineering judgment	Test plan, results, analysis	Ch. 4.3–4.6
Independent acquisition of knowledge	Individual reflection	Ch. 5.3
Standards	IEEE/IETF standards table	Ch. 3.2
Sustainability and societal impact	TWT, BLE, accessibility, SDGs	Ch. 3.7
Risk assessment	Risk register	Ch. 3.7
Security considerations	TLS, SFTP, segmentation, SNMPv3	Ch. 3.7
Integrated system-level engineering	Cross-layer architecture	Ch. 3.5
Project planning and management	Milestones and budget	Ch. 4.8
Individual contribution	Contribution statement and evidence	Front matter + Appendix F